

Quantiles

Quantile $\Rightarrow$ Statistical measure used to divide a set of numerical data into equal size of groups., with each group containing an equal no. of observation.

① Quartile $\Rightarrow$ We divide the data into four equal part (

ex- 100% data (4 part will be -
 $Q_1 = 25\%$, $Q_2 = 50\%$, $Q_3 = 75\%$, $Q_4 = 100\%$
 $\frac{100}{4}$ $\frac{100}{2}$ $\frac{100}{4}$ -

② Quintile $\Rightarrow$ Divide the data into 5 equal parts.

20%	20%	20%	20%	20%
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= 100%

③ Deciles $\Rightarrow$ Divide the data into 10 equal parts.

~~100%~~

10%	10%	10%	10%	10%	10%	10%	10%	10%	10%
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④ Percentile $\Rightarrow$ Divide the data into 100 equal parts.

$$P_L = \frac{P}{100} (N+1)$$

Percentile Location

Percentile

Total no. of observation

Note % $\rightarrow$ $P_L \rightarrow$ It is the location, not the exact no. in the Dataset.

% $\rightarrow$ After Sorting the Data.

data = [78, 82, 84, 88, 91, 93, 94, 96, 98, 99]

What is the percentile of 75%.

$N=10$

$$P_L = \frac{P}{100} (N+1) = \frac{75}{100} (10+1) = \frac{3}{4} (11) = \frac{33}{4} = 8.25$$

$P_L = 8.25 \rightarrow$ Location

$$8^{th} = 96$$

$$9^{th} = 98$$

$$= 8^{th} + 0.25 (9^{th} - 8^{th})$$

$$= 96 + 0.25 (98 - 96)$$

$$= 96 + 0.25 (2)$$

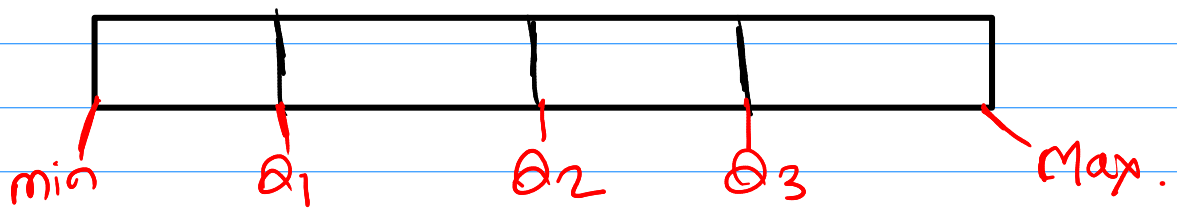
$$= 96.5 \rightarrow 75^{th} \text{ percentile}$$

Outlier Detection

⇒ Box Plot

↳ 5 number Summary

The five no. summary is a descriptive statistics that provides a summary of a dataset. It consists of five values that divide the data into four equal parts (also known as Quartile)



① Min (Lower-boundary) ⇒ It is the smallest possible value in the dataset.
⇒ The value may or may not be available in the Dataset.

$$LB = Q_1 - 1.5 (IQR)$$

② Q_1 ⇒ It is the first Quartile i.e. 25% of the data. $\downarrow$
 $\frac{P}{100} (n+1)$

③ Q_2 = It is the Second Quartile. i.e. 50% of the data.

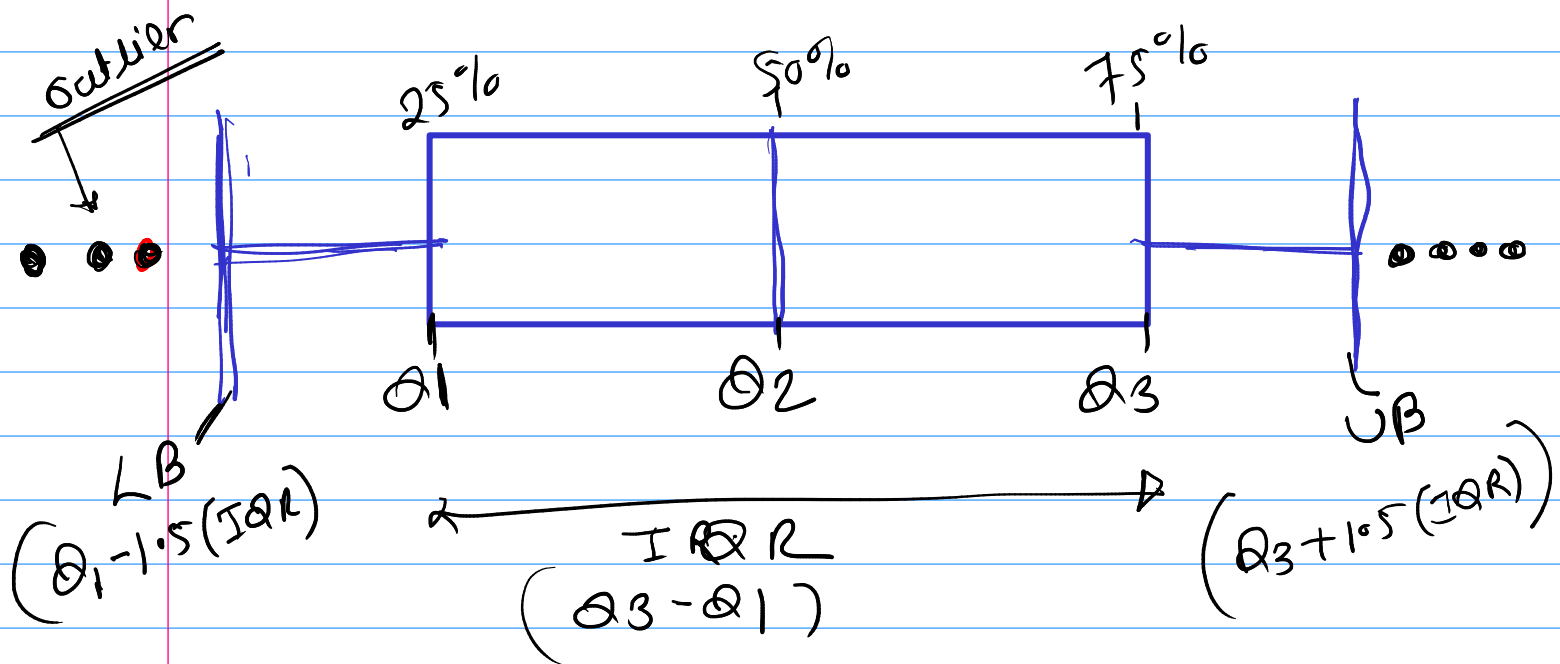
④ $Q_3 \Rightarrow$ It is third Quartile i.e. 75% of the data.

⑤ Max $\rightarrow$ Upper bound $\rightarrow$ It is the possible max value in the dataset.
 $\rightarrow$ It may / may not be available in the dataset.

$$\text{Max} = Q_3 + 1.5(IQR)$$

⑥ IQR $\rightarrow$ Inter Quartile Range :-
It is the range between third Quartile & first Quartile

$$IQR = Q_3 - Q_1$$



6, 213, 241, 260, 281, 290, 314, 321, 350, 1500

1 2 3 4 5 6 7 8 9 10

$$Q_1 = \frac{25}{100}(11) = \frac{11}{4} = 2.75 \Rightarrow 2^{\text{nd}} + 0.75(3^{\text{rd}} - 2^{\text{nd}})$$

$$= 213 + 0.75(241 - 213)$$

$$Q_1 = 234$$

$$Q_2 = \frac{50}{100}(11) = \frac{11}{2} = 5.5 \Rightarrow \left(\frac{290 + 281}{2} \right)$$

$$Q_2 = 285.5$$

$$Q_3 = \frac{75}{100}(11) = 8.25 \rightarrow 8^{\text{th}} + 0.25(9^{\text{th}} - 8^{\text{th}})$$

$$= 321 + 0.25(350 - 321)$$

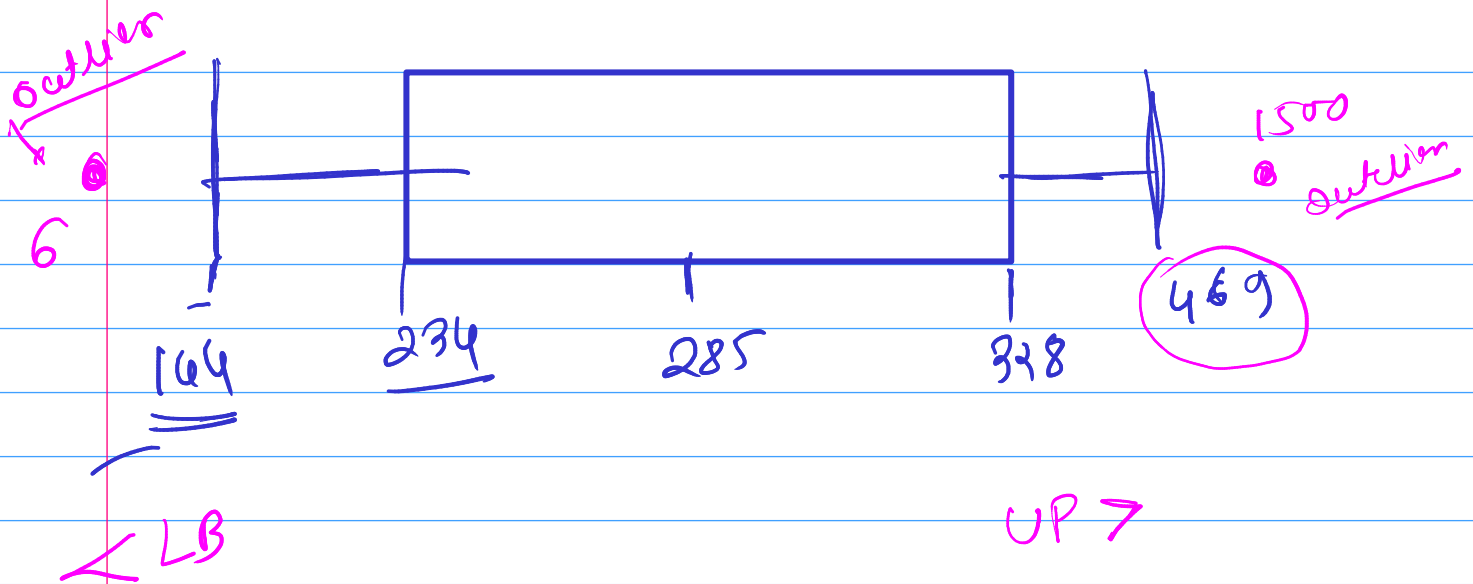
$$Q_3 = 328.25$$

$$IQR = Q_3 - Q_1 = 328.25 - 234$$

$$IQR = 94$$

$$LB = Q_1 - 1.5(IQR) = 234 - 1.5(94) = \underline{\underline{144}}$$

$$UP = Q_3 + 1.5(IQR) = 328.25 + 1.5(94) = \underline{\underline{469}}$$



Probability % → It is the likelihood or the possibility of event happening

Tossing a Coin = {H, T}

$$\text{Probability of event Head} = \left(\frac{\text{event}}{\text{Total outcome}} \right) \frac{1}{2}$$

$$= \frac{1}{2} = \underline{\underline{0.5}}$$

$$P(T) = 1 - P(H)$$

$$= 1 - 0.5 = \boxed{0.5}$$

Type of Variable

Algebraic Variable

$$5 = 10 + x \quad = \underline{5} - \underline{10} = x$$

$$x = -5$$

$$10 = 5 + x$$

Fix Value

Random Variable $\Rightarrow$ not a fix value.

$$X = \{\text{Tossing a coin}\} \quad x = \{H, T\}$$

$$X = \{\text{Throwing a Dice}\} \quad x = \{1, 2, 3, 4, 5, 6\}$$

Random Variable

Discrete (Whole no.)
RV

Continuous RV
(Height, weight)

————— x —————

Distribution $\Rightarrow$

The probability Distribution is a list of all possible outcomes of a Random variable along with their probability value.

Outcomes	H	T
probability	$\frac{1}{2}$	$\frac{1}{2}$

Outcome	1	2	3	4	5	6
prob	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

2 dice

	Dice 1					
Dice 2	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

$$P(2) = \frac{1}{36}$$

$$P(4) = \frac{3}{36}$$

$$P(6) = \frac{5}{36}$$

$$P(3) = \frac{2}{36}$$

$$P(5) = \frac{4}{36}$$

$$P(7) = \frac{6}{36}$$

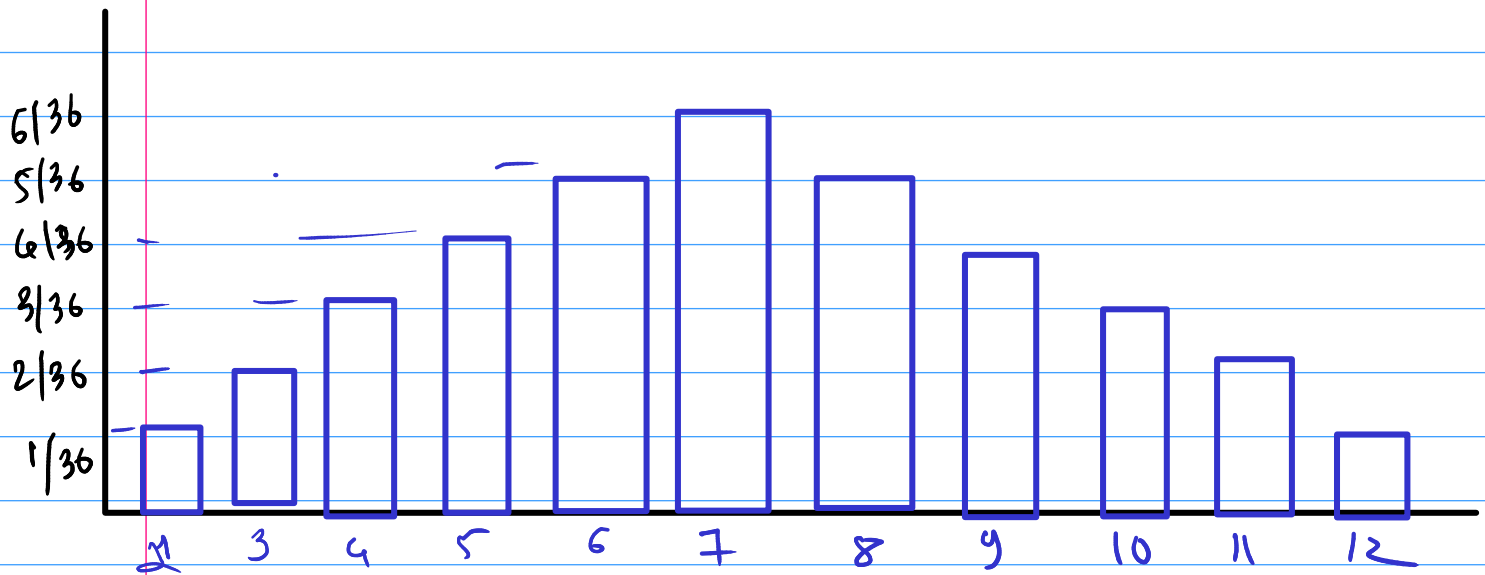
$$P(8) = 5/36$$

$$P(10) = 3/36$$

$$P(12) = 1/36$$

$$P(9) = 4/36$$

$$P(11) = 2/36$$



What if 3 Dice
thrown at a time.